

31338 Network Servers

Lab 4b: Configuring a DHCP server on Windows Server

Aims

1. Configure Windows Server as a DHCP server
2. Test the configuration by using a Linux client

Background Information

This lab follows directly from Lab 4a, but the roles are switched. We will set up the Windows Server as a DHCP server and the CentOS as the DHCP client.

Before you proceed, do not forget to back up your files and configuration from Lab 4a, and make sure you have recorded all the steps and your observations in your engineering journal.

Task 1: Switch the Network

This lab will use a similar configuration as Lab 4a, i.e., use the second network interface (*Ethernet1* on Windows, *ens192* on CentOS).

On CentOS, stop the *kea-dhcp4* service, and change the *ipv4* method to *auto*.

On Windows Server, configure the *Ethernet1* adapter to use a static IP address 10.0.2.2.

Task 2: Install and Configure the DHCP Role

We set up roles via the Server Manager. Configure the Windows Server to be a DHCP server.

1. In Server Manager, go to *Manage -> Add Roles and Features*.
2. From the "Tasks" menu beside Roles and Features, choose "Add Roles and Features"
3. The wizard asks you for:
 - Role-based or feature-based installation
 - Server to install on
 - Roles to install (choose DHCP Server)
 - Whether to install administrator tools
4. In Server Manager, there should now be a new menu item for DHCP. Choose this to configure the server.
5. In Server Manager click *Tools*. In the list, select *DHCP* to run *DHCP Manager*.
6. The DHCP Manager starts in the Microsoft Management Console. There should be three columns. In the left column, click on the server name. You should then see options for IPv4

and IPv6. Right-click on *IPv4* and choose *New Scope*. A scope in Windows Server is equivalent to a range in Linux. A wizard should guide you through adding a new scope. Give a name of this service, e.g., Lab internal. In the IP address range, set the range, e.g., 10.0.2.50 – 10.0.2.100. Optionally set the subnet mask for DHCP clients. You can set the exclusions IP addresses. In the router page, set the default gateway to 10.0.2.1. Ensure your DNS settings consistent to the Windows Server system. Finally, activate the scope.

Task 3: Use CentOS as the DHCP Client

On your CentOS side, reload and activate the network interface ens192. Inspect the IP address using

```
ip addr show ens192
```

or

```
ifconfig ens192
```

to see if the IP address has been automatically allocated. Also, try the ping command to test the connection.

Task 4: Set up a Reserved Address

Windows DHCP server can also reserve IP addresses by MAC addresses. In this test,

1. You can determine the hardware address of the Linux server by running *ifconfig*.
2. On Windows Server, open DHCP Manager and find the scope. If you expand the scope menu option, there should be a sub-option called "Reservations".
3. Right-click on Reservations, then add a new reservation for this mac address.
4. set the IP address to 10.0.2.30. Notice that the MAC address must be entered in the format with dashes between each pair of hex digits rather than colons. If you get an error message that "The unique identifier you have entered may not be correct", then this likely means your MAC address is not in the correct format.
5. Click Add. Note that this screen doesn't give an acknowledgement! Use Close button to continue.

Reload and activate the network interface ens192 in CentOS, then observe the IP address.